Governance Under Adversarial Strategy Injection in Sequential Commons

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Abstract

We study governance under adversarial strategy injection in sequential commons through two linked studies. The first study uses a minimal Fishery Commons environment to compare central governance signals under repeated strategy turnover and held-out regime shifts. A matched mutation-versus-live-LLM baseline shows that monitoring with sanctions reduces held-out collapse relative to no governance for both injector types. A broader top-down sweep over partner mixes and pressure levels then identifies adaptive quotas as the strongest overall signal, and a learned-policy self-play validation in the medium tier recovers the same ordering: adaptive quotas outperform no governance on held-out collapse, stock, and welfare while also exceeding the other governed conditions. The second study extends the same invasion protocol to Harvest Commons and shifts the governance question from signal design to architecture design, with the manuscript's main comparison restricted to top-down-only versus hybrid control. In the high-power search-over-mutations matrix, hybrid ranks first in all eight decision cells against top-down-only control under a failure-then-patch-health-then-welfare rule, improving patch health and usually reducing neighborhood overharvest, with a moderate average welfare cost. A narrower medium-tier PPO validation yields concordant evidence: hybrid self-play improves patch health and lowers local aggression and neighborhood overharvest relative to top-down-only control, with only a small welfare cost. Together, the studies provide a two-level evaluation protocol that first identifies effective central governance signals in a controlled commons and then tests a narrower governance-architecture comparison in a second substrate under the same turnover pressure.

1 Introduction

Classic commons theory predicts over-extraction under open access and highlights institutional design as the key lever for sustainable use [2, 7, 11]. In machine learning, sequential social dilemma research has made the same point in a different language: cooperation is a property of policies interacting over time in stateful environments, so evaluation must account for long horizons, partial observability, and held-out social situations rather than one-shot payoffs [1, 3–5]. Recent LLM-agent commons work shows that cooperative behavior can emerge in simulated societies, but those findings remain fragile under changes in prompts, partner populations, and environmental conditions [9, 10]. What remains weakly characterized is whether governance itself stays effective once new exploitative strategies keep entering the population.

We frame this problem as *governance under adversarial strategy injection*. The central question extends beyond a single fishery result: *how can governance stabilize cooperation under repeated strategic turnover in sequential commons, and how do the relevant design choices change when governance moves from fixed central signals to broader governance architectures?* To address this question, the paper is organized as two linked studies. The first study uses a minimal Fishery Commons setting to identify which *central governance signals* remain effective under invasion pressure. It begins with a matched mutation-versus-live-LLM baseline, then widens into a top-down comparison over monitoring with sanctions, adaptive quotas, and temporary closures, and finally evaluates the resulting ranking with a learned-policy self-play validation in the same medium tier. The second study moves to Harvest Commons and asks a downstream question: once useful central signals have been identified, how should governance itself be organized? Earlier Harvest experiments also considered none and bottom-up-only conditions, but the decision-critical evidence in the manuscript is the narrower top-down-only versus hybrid comparison, with a learned-policy check reported only as supporting validation.

This design reflects two considerations. Commons research has long shown that communication, collective choice, and locally enforced rules can materially change outcomes even without full central command [8, 12]. At the same time, benchmark work in multi-agent learning increasingly argues that robust conclusions should survive transfer across substrates and social situations, rather than depending on a single tuned environment [3, 5]. The Fishery Commons study therefore isolates the signal-design question, whereas the Harvest Commons study widens the lens to governance architecture. Taken together, the two studies address the same overarching problem at adjacent levels: which central interventions work, and what changes when governance is no longer only a hard-coded central rule.

The paper introduces a two-level invasion-stability protocol for sequential commons. First, it uses repeated adversarial strategy turnover, held-out regime evaluation, run-level uncertainty, and learned-policy validation to compare central governance signals in a minimal renewable commons. Second, it extends the same pressure test to a new commons substrate and evaluates a narrower governance-architecture comparison between top-down-only and hybrid control, while situating that comparison within a broader experimental setting that also includes none and bottom-up-only conditions. The resulting evidence supports concrete empirical claims while remaining relevant to decentralized AI settings in which governance is part of the system design.

2 Methods

Both studies use the same core invasion protocol. A population of strategies is evaluated over generations, weaker strategies are removed, and new strategies are injected through a specified generator. Selection is performed in train conditions and reported robustness is measured in held-out test conditions. This separates adaptation pressure from generalization assessment and makes governance comparisons about out-of-distribution stability rather than only in-sample performance.

2.1 First study: Fishery Commons and central governance signals

The first study uses a minimal renewable fishery commons with a shared stock, per-agent harvest actions, regeneration, and collapse dynamics. The main invasion runs use interpretable threshold strategies, and governance changes extraction pressure directly through monitoring, quota clipping, sanctions, adaptive quotas, or temporary closures. The environment logs requested harvest, realized harvest, quota-clipped harvest, repeat-offender rate, closure activity, and stock-recovery lag so that governance can be analyzed mechanistically rather than only through final collapse outcomes.

The Fishery Commons study has two invasion layers plus one learned-policy validation. The first invasion layer is a matched baseline that compares no governance, monitoring alone, and monitoring with sanctions across two injector types: mutation and live JSON-based LLM injection through Ollama with the ‘qwen2.5:3b-instruct‘ model. This baseline is run on the easy and medium fishery tiers. The second invasion layer is a broader top-down sweep, restricted to the medium tier, that compares no governance, monitoring with sanctions, adaptive quotas, and temporary closures under three non-LLM injector families: mutation, an adversarial heuristic injector, and search over mutations. The sweep also varies partner composition between balanced and adversarial-heavy populations, and it varies adversarial pressure between 0.3 and 0.5. This layer identifies which central governance signals remain effective once the setting becomes more hostile.

The learned-policy validation asks a narrower question: does the resulting governance ranking survive outside the hand-specified threshold family? For this check, agents share a parameter-shared PPO policy with a small multilayer perceptron, observe normalized stock and normalized timestep, and choose among 11 harvest bins. Training uses the same fishery train regime as the matched baseline (‘regen_rate = 2.0‘, ‘obs_noise_std = 6.0‘) for 250,000 environment steps per seed. Final checkpoint selection uses train-regime evaluation only, and held-out reporting uses the fixed ‘medium_v1‘ benchmark pack. A direct five-seed confirmatory comparison evaluates no governance against adaptive quotas, and a matched five-seed governance follow-up compares monitoring with sanctions, adaptive quotas, and temporary closures under the same PPO protocol. This evaluation serves as a policy-family robustness check on the top-down ranking rather than as an additional invasion matrix. The shared Fishery invasion defaults for the matched baseline are 15 generations, 32 train and 32 test seeds per generation, replacement fraction 0.2, and adversarial pressure 0.3.

2.2 Second study: Harvest Commons and governance architecture

The second study introduces Harvest Commons, a second renewable commons substrate built around local patches rather than a single global stock. It uses the same invasion logic as the first study rather than only a fixed-composition matrix. A population of six parameterized Harvest strategies is evaluated, ranked, and partially replaced over generations. The evolving strategies control local harvest fractions, restraint signals, credit requests, credit offers, reciprocity weights, and cap-compliance margins. Communication and side-payments remain condition-controlled by the environment, so the same evolving strategy family can be evaluated under different governance architectures.

Across the broader Harvest study, four governance conditions are defined: none, top-down only, bottom-up only, and hybrid. Top-down only uses a fixed government agent that observes patch stress and local aggressive neighborhoods, then applies targeted caps. Bottom-up only enables local communication and side-payments without central intervention. Hybrid combines the fixed government agent with local communication and credit transfers. The earliest Harvest evidence is a fixed-composition pilot over that broader architecture set, retained only as precursor evidence. A subsequent invasion pass with lighter non-LLM injectors suggested that hybrid governance warranted closer analysis. The principal Harvest comparison in the manuscript narrows to a stronger search-over-mutations comparison between top-down-only and hybrid, where the invader repeatedly proposes the most damaging child among multiple mutated candidates.

The strengthened Harvest cycle begins with a pilot matrix over the medium and hard tiers, the balanced and adversarial-heavy partner mixes, top-down only versus hybrid governance, mutation versus search over mutations, and adversarial pressures 0.3 and 0.5. The decision-critical Stage C matrix then fixes the injector to search over mutations and reruns those same medium and hard, balanced and adversarial-heavy, top-down and hybrid cells at higher power. Each decision cell is one ‘(tier, partner mix, pressure)‘ row in that Stage C matrix, and each cell is run for 15 generations with 32 train and 32 test seeds per generation. In addition to patch health and welfare, the Harvest Commons study reports local mechanism metrics including local aggression, neighborhood overharvest, capped-action rate, targeted-agent fraction, and prevented harvest. These metrics are necessary because a governance architecture can change *where* exploitation occurs, not only the global average outcome.

The second study also includes a narrow learned-policy validation. For this check, agents share a parameter-shared PPO policy with separate communication and action heads. Communication is chosen from local observations, while harvest and credit-offer decisions condition on local observations plus inbox summaries from neighboring agents. The validation is restricted to the ‘medium_h1‘ tier and compares top-down-only against hybrid governance over five training seeds, each run for 100,000 environment steps. Training reward combines welfare with a patch-health bonus and mild penalties on local aggression and neighborhood overharvest. Held-out reporting uses the fixed ‘medium_h1‘ benchmark pack. This validation is reported as supporting evidence for the architecture comparison rather than as a replacement for the main Stage C invasion matrix.

2.3 Statistical reporting

The primary matched-baseline outcome in the Fishery Commons study is held-out collapse reduction, reported as the difference between no governance and monitoring with sanctions. Supporting metrics are held-out stock, time-to-collapse, and per-regime survival. Confidence intervals for the matched baseline are run-level bootstrap intervals with 5,000 resamples. The broader Fishery Commons sweep uses the same run summaries but is interpreted as a ranking study: adaptive quota, temporary closure, and monitoring with sanctions are compared on collapse first and welfare second. The learned-policy validation reports the same held-out collapse, stock, welfare, and per-regime survival summaries over five independent PPO training seeds.

The primary comparison in the Harvest Commons study is the paired difference between hybrid and top-down-only governance at matched run identifiers within each tier, partner mix, and adversarial-pressure cell. This comparison isolates whether local communication and credit sharing add anything once central intervention is already present. Confidence intervals are again estimated by bootstrap over paired deltas. The manuscript also reports best-condition counts over the Stage C decision cells. Those counts use the same ranking rule as the archived summary artifacts: lower garden failure first, higher patch health second, and higher welfare third. Accordingly, references to a condition “winning” a decision cell denote first rank under that lexicographic rule rather than superiority on every reported metric. In the high-power Stage

C matrix, garden failure is never the discriminating outcome, so the main Harvest Commons evidence is carried by patch health, welfare, local aggression, and neighborhood overharvest. The supporting Harvest PPO validation reports held-out means over five independent training seeds in ‘medium_h1‘; because garden failure is saturated there as well, interpretation again focuses on patch health, welfare, local aggression, and neighborhood overharvest.

Formal invasion dynamics and the saturation identifiability condition are provided in supplementary materials and source files. In the empirical pipeline, the reported quantities are estimated from run-level summaries and archived result artifacts to avoid post-hoc threshold drift.

3 Results

3.1 First study: central governance signals in Fishery Commons

The matched Fishery Commons baseline shows that even a minimal central intervention survives repeated injection pressure. Under five paired high-power runs in the medium tier, monitoring with sanctions reduces held-out collapse relative to no governance for both injector types. For mutation, collapse reduction is 0.1099 with 95% confidence interval $[0.0310, 0.1896]$. For live JSON-based LLM injection, collapse reduction is 0.0835 with 95% confidence interval $[0.0154, 0.1509]$. Mean held-out stock also rises substantially, by 23.35 for mutation and 22.40 for live LLM injection. These matched results establish that the invasion-stability protocol is not an artifact of one injector family and that a lightweight central signal already improves robustness.

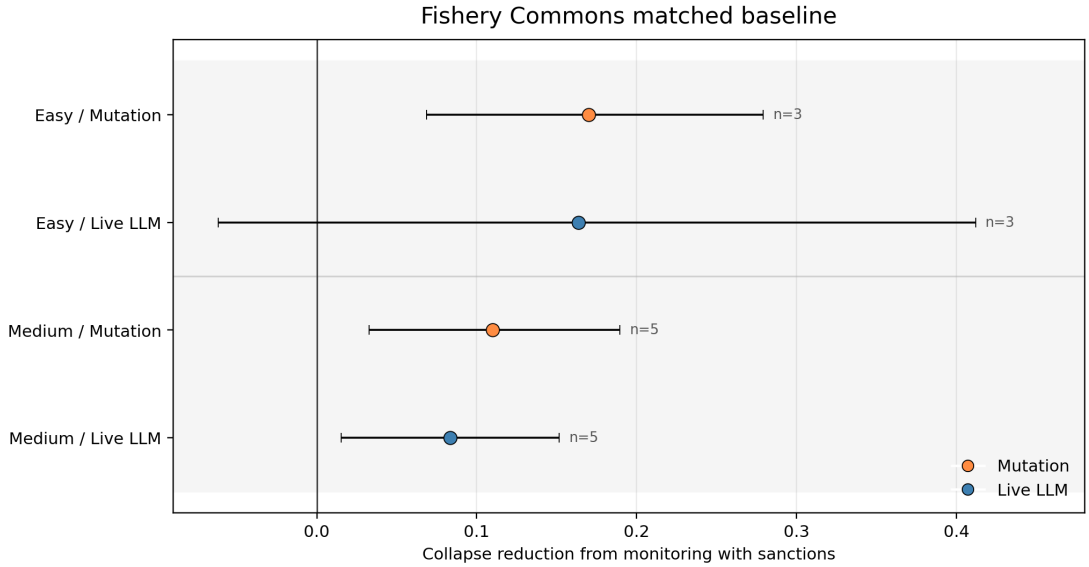

Figure 1: Fishery Commons matched baseline. Monitoring with sanctions reduces held-out collapse relative to no governance for both mutation and live LLM injection. Dots show paired-run means and whiskers show 95% confidence intervals.

The broader top-down sweep in the medium fishery tier sharpens the central-governance question. Across balanced and adversarial-heavy partner mixes, pressure levels 0.3 and 0.5, and three non-LLM injector families, adaptive quotas are the most frequent winner in every injector block. They win all four scenarios for mutation, all four for the adversarial heuristic injector, and all four for search over mutations. This pattern arises because adaptive quotas and temporary closures tie on collapse reduction in most cells, but adaptive quotas usually retain more welfare outside the most hostile partner mixes.

Table 1: Fishery Commons medium-tier top-down comparison. Collapse reductions are measured relative to no governance, and the final column reports the welfare advantage of adaptive quotas over temporary closures.

Injector	Sanctions	Adaptive quota	Closure	Adaptive quota minus closure welfare
Mutation	0.3737	0.9167	0.9167	2408.81
Adversarial heuristic	0.0208	0.9167	0.9167	2211.61
Search over mutations	0.2617	0.9167	0.9167	2723.02

The welfare comparison is heterogeneous across partner mixes. In balanced partner mixes, adaptive quotas dominate closures on welfare by large margins: 4792.55 for mutation, 5202.70 for the adversarial heuristic injector, and 5367.27 for search over mutations at pressure 0.3. In adversarial-heavy partner mixes, that advantage shrinks or reverses: the same adaptive-minus-closure welfare difference is −215.62, −779.48, and −71.33 at pressure 0.3 for the three injectors. These results indicate that adaptive quotas are the strongest overall signal, while closures remain a credible fallback when ecological protection is prioritized over welfare retention.

Mechanism logging explains why monitoring with sanctions is weaker than the more forceful top-down alternatives. Mean quota-clipped harvest under monitoring with sanctions is 8.53 for mutation, 18.32 for the adversarial heuristic injector, and 9.61 for search over mutations, while mean repeat-offender rates remain 0.3032, 0.5568, and 0.3458. Monitoring with sanctions therefore reduces harm without fully suppressing aggressive extraction, whereas adaptive quotas and temporary closures push collapse to zero in nearly every cell of the first Fishery Commons study.

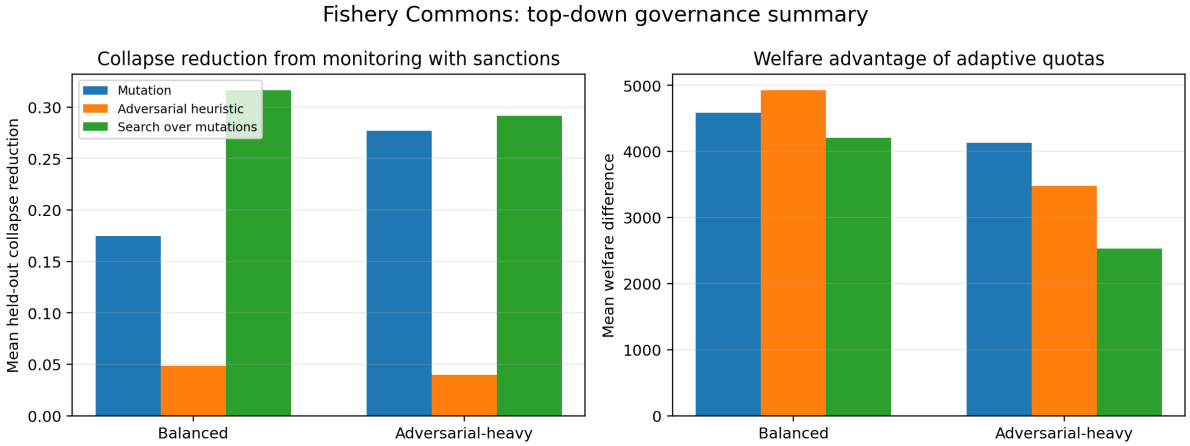

Figure 2: Fishery Commons top-down summary. Left: mean held-out collapse reduction from monitoring with sanctions relative to no governance. Right: mean welfare advantage of adaptive quotas over temporary closures. Adaptive quotas dominate in balanced settings but lose some advantage in adversarial-heavy mixes.

A learned-policy validation reaches the same conclusion. Under five PPO self-play training runs on the held-out ‘medium_v1‘ pack, adaptive quotas reduce mean collapse from 0.8141 under no governance to 0.0, raise mean stock from 97.56 to 186.05, and raise mean welfare from 4770.20 to 9902.35. A matched governance follow-up under the same PPO protocol also ranks adaptive quotas above the other governed conditions: monitoring with sanctions reaches held-out collapse 0.0016, stock 126.52, and welfare 6800.78, while temporary closures reach collapse 0.0, stock 124.05, and welfare 4719.58. These results indicate that the first-study ranking is not confined to hand-written threshold strategies. A learned-policy family recovers adaptive quotas as the strongest medium-tier governance signal.

3.2 Second study: governance architecture in Harvest Commons

The second study applies the same invasion protocol used in Fishery Commons rather than relying only on a fixed-composition comparison. Precursor experiments also included none and bottom-up-only conditions, but the decision-critical evidence reported here is restricted to top-down-only versus hybrid. The fixed-composition Harvest matrix and the initial invasion pass with lighter non-LLM injectors both identified hybrid governance as the stronger architecture in this narrowed comparison. The principal Harvest evidence comes from the search-over-mutations cycle. Its Stage B pilot showed that hybrid governance remained competitive under a stronger invader: hybrid ranked first in 13 of 16 pilot cells, whereas top-down-only ranked first in the remaining 3. The main Harvest Commons evidence therefore comes from the Stage C high-power search-over-mutations matrix, which restricts attention to the decision-critical rows in the medium and hard tiers, the balanced and adversarial-heavy partner mixes, top-down-only versus hybrid governance, and pressures 0.3 and 0.5.

Table 2: Harvest Commons high-power search-over-mutations summary for the top-down-only versus hybrid comparison. Best-condition counts use the manuscript's Stage C ranking rule: lower garden failure, then higher patch health, then higher welfare. Positive deltas mean hybrid governance exceeds top-down-only control; negative values are improvements for lower-is-better metrics such as local aggression and neighborhood overharvest.

Summary quantity	Value
Hybrid top-ranked decision cells	8 / 8
Top-down-only top-ranked decision cells	0 / 8
Mean patch-health delta	+0.0438
Mean welfare delta	-0.0918
Mean local-aggression delta	-0.0037
Mean neighborhood-overharvest delta	-0.0646

In that stronger high-power matrix, hybrid governance ranks first in all 8 decision cells under the Stage C failure-then-patch-health-then-welfare rule. The central inferential question is whether hybrid continues to outperform top-down-only control once the invader actively searches for more damaging children. The Stage C results show that it does, although the associated trade-off is sharper than in earlier Harvest experiments.

The aggregated Stage C view makes that trade-off readable. Patch health rises in every single decision cell, and every patch-health confidence interval remains above zero. In the hard tier with an adversarial-heavy population, the paired patch-health delta is $+0.0157$, with a 95% confidence interval from 0.0027 to 0.0264, while the welfare delta is -0.0883, with a 95% confidence interval from -0.1668 to -0.0119. In the hard tier with a balanced population at pressure 0.3, hybrid improves patch health by $+0.1356$, with a 95% confidence interval from 0.0303 to 0.2722, but welfare falls by 0.2306, with a 95% confidence interval from -0.3922 to -0.0643.

Medium-tier effects are more heterogeneous, but remain directionally consistent. In the adversarial-heavy block at pressure 0.3, patch health rises by $+0.0474$, with a 95% confidence interval from 0.0140 to 0.1019, while welfare overlaps a modest cost. In the balanced block at pressure 0.3, both patch health and welfare favor hybrid in point estimates, although the welfare interval crosses zero. The largest welfare penalty appears in the medium balanced block at pressure 0.5, where patch health still rises by $+0.0438$, with a 95% confidence interval from 0.0052 to 0.0863, but welfare falls by 0.2866, with a 95% confidence interval from -0.5390 to -0.0702. These results do not imply universal metric dominance or zero welfare cost for hybrid governance. The narrower conclusion is that, under a stronger search-based invader, hybrid governance preserves the commons more effectively than top-down-only control under the manuscript's ranking rule, while the welfare cost depends strongly on social composition and pressure.

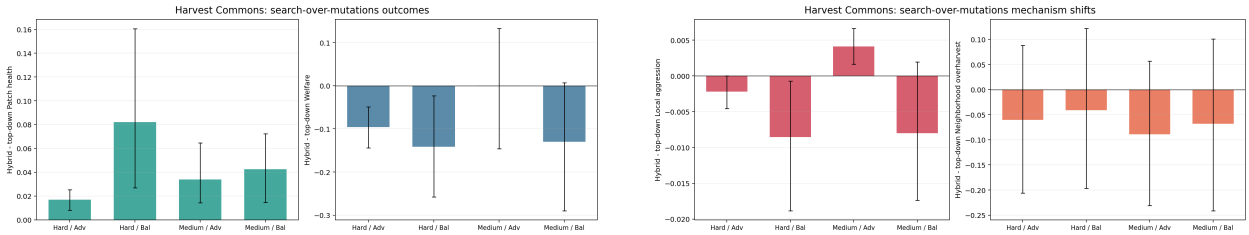

Figure 3: Harvest Commons search-over-mutations summary. Left: hybrid governance improves patch health in every block, while welfare depends on social mix and pressure. Right: negative values are improvements for lower-is-better metrics, showing that hybrid governance lowers local aggression on average and usually reduces neighborhood overharvest.

These mechanism metrics explain why the failure metric is uninformative in the Harvest Commons study. Across all 8 high-power rows, the paired garden-failure delta is exactly zero, so governance separation happens before outright garden failure. Hybrid governance matters because it shifts local behavior: averaged over the full Stage C matrix, local aggression falls by 0.0037 and neighborhood overharvest falls by 0.0646 relative to top-down-only control. The stronger result is therefore behavioral and ecological rather than catastrophic, and the 8-of-8 ranking result should be read through those mechanism margins rather than as a claim of metric-by-metric domination.

A narrow learned-policy validation yields concordant evidence. Under five PPO self-play training runs on the held-out ‘medium_h1‘ pack, hybrid raises mean patch health from 15.5673 under top-down-only control to 15.8064, lowers mean max local aggression from 0.6 to 0.4, and lowers mean neighborhood overharvest from 9.0 to 7.65. Mean welfare falls only slightly, from 16.3757 to 16.2898. Because garden failure is again saturated at zero in this medium-tier RL setting, the learned-policy check is informative only on the same local mechanism margins that matter in the invasion matrix. It should therefore be read as supporting evidence for the architecture ranking, not as a substitute for the stronger search-over-mutations result.

Table 3: Harvest Commons PPO validation in ‘medium_h1‘. Lower values are better for local aggression and neighborhood overharvest. The table provides supporting evidence, whereas the main Study 2 claim remains anchored in the high-power search-over-mutations matrix.

Governance	Patch health	Welfare	Max local aggression	Neighborhood overharvest
Top-down only	15.5673	16.3757	0.6	9.0
Hybrid	15.8064	16.2898	0.4	7.65

Taken together, the two studies provide a layered invasion result. In the controlled fishery setting, central governance signals can be ranked, and adaptive quotas emerge as the strongest overall top-down signal under invasion pressure. In the second substrate, once governance is allowed to become architectural rather than only signal-level, the narrower top-down-only versus hybrid comparison shows that hybrid produces better local ecological control under a stronger search-based invader, although the welfare effect depends on both social composition and task difficulty.

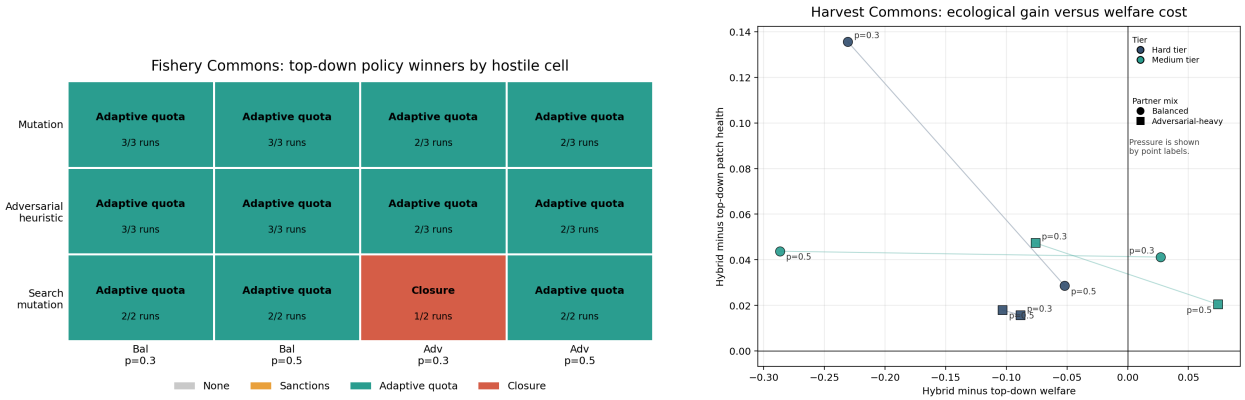

Figure 4: Comparison of Fishery Commons and Harvest Commons. The left panel summarizes which top-down policy most often ranks first in the fishery setting. The right panel shows the Harvest Commons trade-off between ecological gain and welfare cost when hybrid governance is compared with top-down-only control; point color marks tier, marker shape marks partner mix, and labels indicate adversarial pressure.

4 Discussion

The results support two related claims. First, central governance signals can be ranked under adversarial strategy injection in a minimal sequential commons. The Fishery Commons study shows that monitoring with sanctions improves held-out collapse under both mutation and live LLM injection, thereby establishing the minimal robustness result. The broader signal sweep then shows that stronger top-down interventions are not interchangeable: adaptive quotas are the best overall policy in medium-tier fishery runs, while temporary closures remain a credible fallback under adversarial-heavy partner mixes. A learned-policy PPO validation in the same tier reaches the same conclusion, so that ranking is not confined to the hand-specified threshold family. This distinction is analytically useful because it separates light-touch enforcement from stronger adaptive intervention.

Second, governance should not be treated only as a purely central instrument. The Harvest Commons study addresses the broader question of governance architecture. Earlier experimental stages included none and bottom-up-only conditions, but the decision-critical comparison reported here narrows to hybrid versus top-down-only control. The central empirical finding is that hybrid governance improves local ecological control relative to top-down-only control, although this improvement carries a measurable welfare cost in several cells. Earlier Harvest pilots and lighter invasion runs suggested this pattern, and the stronger search-over-mutations stress test provides the principal support for it. In the high-power Harvest matrix, hybrid ranks first in all 8 decision cells under the failure-then-patch-health-then-welfare rule and improves patch health in every tier-by-partner block, but several rows show a clearer welfare cost than before. A narrower Harvest PPO validation in the medium tier yields concordant evidence, which indicates that the architecture ranking is not confined to the hand-designed Harvest strategy family. The RL evidence remains secondary because it is medium-tier, fixed-governance, and does not constitute an invasion benchmark. An evaluation restricted to a single collapse statistic would obscure this trade-off. These findings support the move from a single global stock to a second substrate with local interactions and communication.

This framing also explains why the two studies belong in one paper. The Fishery Commons study isolates which central signals work in a highly controlled environment. The Harvest Commons study then asks whether governance remains purely central once local coordination, communication, and targeted intervention are allowed to matter, while evaluating that question through a narrower top-down-only versus hybrid stress test. Commons experiments have long suggested that communication and collective choice can improve outcomes even without a single command-and-control institution [8, 12]. The results reported here extend that intuition to adversarially changing agent populations: once strategy turnover is explicit, governance must be evaluated both as a set of central signals and as a broader architecture for decentralized systems.

The implications extend beyond the specific substrates studied here. More broadly, the paper concerns institution design for self-interested multi-agent systems under uncertainty and turnover. The same protocol

can serve as a testbed for shared compute budgets, energy systems, congestion control, spectrum allocation, or decentralized markets, where new behaviors keep entering and governance must balance resilience against performance. Recent generative-society work argues for realistic evaluation under diverse partners and conditions [6, 9, 10]; the present results add a more specific lesson: governance should be evaluated both at the level of signal choice and at the level of governance architecture, and stronger invaders can tighten without necessarily overturning those conclusions even when the principal architecture comparison is deliberately narrow and decision-focused.

5 Limitations

The two studies answer adjacent questions, not identical ones. The first study is a controlled signal-ranking study in Fishery Commons, while the second study is a governance-architecture extension in Harvest Commons. The second study shares the same invasion loop and replacement logic as the first, which resolves the earlier asymmetry, but it still should not be read as a strict replication in another substrate because the governance question itself is broader.

The LLM comparison appears only in the Fishery Commons study as main-paper evidence. These results are sufficient to show that minimal central governance survives both mutation and live JSON-based strategy generation, but they do not establish the same result for Harvest Commons. A live Harvest LLM path was implemented and cleared an initial reliability screen, but the matched narrow Harvest pilot still failed the manuscript's evidence gate because schema completeness remained too uneven: the mean effective-LLM fraction stayed at 0.7311 and unrepaired fallback averaged 0.2689, with missing keys as the dominant failure mode. Accordingly, Harvest LLM evidence is treated only as diagnostic, and the main Study 2 claim remains anchored in the stronger non-LLM search-over-mutations evidence.

The learned-policy extensions remain narrow. In Fishery Commons, the PPO validation is limited to the medium tier and fixed governance conditions. In Harvest Commons, a medium-tier PPO self-play validation supports the `hybrid` > `top_down_only` direction, but it still uses fixed governance conditions, shaped rewards, and held-out evaluation in only one tier. These checks are sufficient to show that the leading Fishery and Harvest rankings are not artifacts of one hand-specified policy family, but they are not yet learned-policy invasion benchmarks and they do not replace the main mutation or search-over-mutations evidence.

Hard settings function primarily as stress tests. In fishery, once collapse saturates at 1.0 and survival at 0.0 across conditions, governance separation becomes non-identifiable under floor effects. In Harvest Commons, the harder tiers remain informative, but garden failure itself does not separate conditions in the high-power rows; the main signal comes from patch health, welfare, and local behavioral metrics.

Finally, both substrates remain intentionally stylized. This stylization aids identification, but it limits external realism. Market feedbacks, richer heterogeneity, richer language interaction, and explicit institutional bargaining are natural extensions once the current governance comparisons are fixed.

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